GR8

Team 8 – Gautam, Sarvesh

BrainBook High-Fi Prototype

Value Proposition

A Stimulating, Personalized Guide for the Modern Age

Problem/Solution

Problem: Current AI tutoring often collapses the learning process into “ask to get answer,” which reduces student agency and skips the step where students externalize and debug their own reasoning.

Solution: We solve this with a neutral, adaptive AI textbook that gives small, digestible hints and then asks targeted follow-up questions so the student actively builds the explanation themselves.

Table of Contents

01 High Level Summary

02 Justifications

03 Major Revisions

04 Figma Link

High Level Summary

Violations

78 Violations
Across Three Evaluations

In all three evaluations, violations were spread very evenly across all 10 categories. However the most common violations were:

- **H1: Visibility of System Status (12)**
  - Commonly due to the system not telling the user that certain elements were completed (like the learning loop) or reset
- **H4: Consistency & Standards (12)**
  - Commonly due to using the same Orange Pill design to represent different types of inputs
- **H6: Recognition not Recall (11)**
  - Commonly due to a lack of placeholders

Severity Breakdown

- Level 4: **11 Violations**
- Level 3: **28 Violations**
- Level 2: **34 Violations**
- Level 1: **5 Violations**

This suggests that most of our violations where non-cosmetic usability errors that had a small-to-medium impact on user experience, in addition to a moderate amount of severe usability violations

- Most **severe** violations concerned users being unable to access previous or new information
- Most **major** violations concerned the system not informing the user of state change
- Many **minor** violations concern design implications, unclear labels, or missing explanations, among other things

UI Revisions

8 Major UI Changes
Made Across the App

These changes were inspired mostly be level 3 & 4 violations that required immediate action

- Some level 3 & 4 violations fundamentally misunderstood our app and our Wizard-of-Oz techniques (tapping to replicate writing, circling, etc.), so they were ignored

Some easy changes inspired by level 1 & 2 violations were made as well

Justification

Level 4 Violations

- **Tapping anywhere advances the conversation:** We elected to *ignore* this violation, since this behavior was specifically chosen for several reasons:
    - Tapping is how the prototype simulates writing
    - Since there isn't an actual AI providing responses, we had to make some tradeoffs on how to represent AI-Human interaction
        - Figma also limits what we can do
    - We decided to give the user full control over showing both the AI and User responses for transparency
    - Since the user can potentially write (and tap) anywhere on their side, and they also control the AI, it made sense to progress the conversation by allowing tapping anywhere
- **The Learning Loop doesn't strongly indicate its completion:** we both made the final human response more obvious that it was done, and created a modal that pops up to explicitly signal the end of the session
    - These changes more closely mirror how a human would actually "end" the session (although these sessions can't end), and the pop-up is the strongest possible indication of completion
- **The system doesn't show users that they can scroll right to the User Space:** to address this, we included the AI/User Space scrolling in the onboarding manual (to explicitly and clearly inform the user) and also provided a hint telling the user to scroll right after the first AI response (to remind the user that they can scroll)

Level 4 Violations

- **The Open button is always clickable, even the fields are not filled in:** we chose to *ignore* this violation:
    - Our prototype models a physical device (think Microsoft Surface Duo)
    - Thus, it's always possible for the user to open the physical device, even if they haven't filled in all three inputs
        - In this case, the device would have to show an error screen
    - The Open button is designed to mimic this behavior insofar that the Open button is akin to opening the foldable
    - So, we chose to show an error screen since opening is always possible
- **The Favorite Learning Tools section shows blank lines with no hints:** this fix was pretty simple: add placeholders over each blank line that suggests the types of things that users can enter
    - This intuitively and immediately suggests what sort of elements the user can write in
- **There's no visual connection between associated AI questions and User responses:** designing a fix for this was harder because of the way that our product is designed
    - Ultimately, we went with a simple solution: number associated AI questions and User responses
    - While it's not sophisticated, it easily and immediately tells the user what to expect
- **Non-Obvious Text Entry Spot in My Response:** we agreed with this one, so we added a simple placeholder in My Responses when no text has been added yet informing the user that they may write there – clean and simple!

Level 4 Violations

- **When the AI Space fills up, it suddenly and without warning clears the page:** to fix this, we did two things:
  - Added a warning that the page would reset soon
    - Upfront transparency about what will happen avoids confusion
  - Added navigation buttons between the old and new page to allow the user to see their entire session history
    - The button also helps inform the user about what happened (that a new, indexable page was added after the first page), avoiding confusion

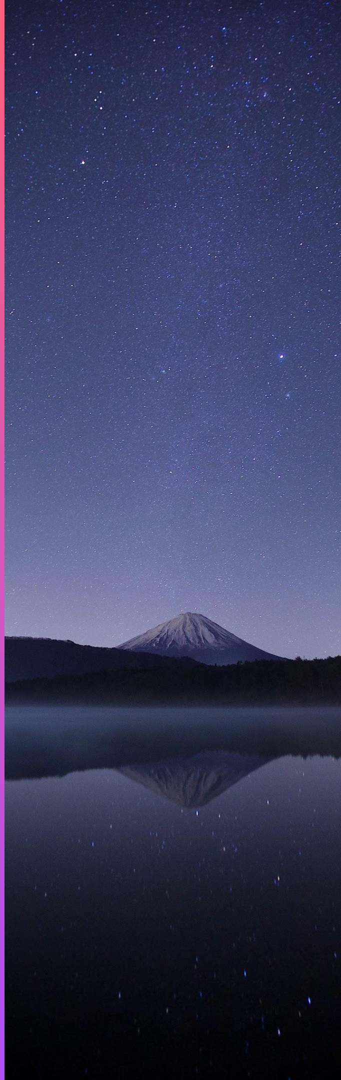
Level 3 Violations

- **The concept of "AI Space" and "User Space" is jargon-like:** we agree, so we renamed AI Space/User Space to "AI Feedback"/"My Responses" – much more personalized and self-obvious
- **Clicking X immediately exits the session with no confirmation/information:** it's understandable how this can cause confusion and worry, so we added a popup modal that confirms if the user wants to actually exit or stay (while informing that the session was saved)
  - This not only informs the user but returns gives them a safety layer on exiting – increased control!
- **The user must remember what the AI said while scrolling between the AI Space and the User Space:** that's the whole point! The user needs to understand and *remember* what the AI says (we chose to *ignore* this)
- **There is no in-app help explaining how to use the two-screen swiping model:** we addressed this by explaining it in both the initial main page (through a hint) and through our onboarding guide
- **On the Closed Page, there is no indication of which prompts have been completed:** we *ignored* this one, because it's self-evident that a prompt is completed when it has text in it
  - Just to be safe, we explained this in our onboarding guide to reduce confusion
- **During the learning loop, there is no progress indicator showing how far into the conversation the user is:** we *ignored* this because sessions have no set limit, so there's no way to measure "progress"
- **During the learning loop, there is no way to go back and review a previous AI or User response once it has been scrolled past or cleared:** this is a critical flaw of our design, so we added navigation buttons to go back and forth

Level 3 Violations

- **The Closed Page has no placeholders:** we added placeholders that tell the user where/what to write
  - The user is immediately shown how to fill out the page
- **When mid-scroll between AI Space and User Space, content from both panels is visible simultaneously, creating clutter:** the user has power to scroll past this mid-scroll view (and why wouldn't they?), so we elected to *ignore* this. There's not much we can do within Figma to address this either
- **The orange pill shape is used for two completely different interactive elements across the prototype:** on the Closed page, we removed the orange pill and replaced it with an underline
  - This both solves the inconsistency of the orange pill's purpose and creates consistency with the rest of the page, reducing confusion across-the-board
- **Once a Favorite Learning Tool has been added on the Settings page, there is no visible way to remove it:** we didn't even think about this! To address this, we added an X button next to each individual tool to give the user the option to remove the learning tool
  - The X button is intuitive and obvious to all users
- **After closing Settings with the X button, there is no confirmation of settings saving:** we didn't consider this either, so we added a short modal that informs the user that their settings were saved
- **User-entered text is displayed in red throughout the prototype across all tasks and screens:** this is *wrong* – user text is distinctly orange and not red, which does not carry the same connotations

Changes

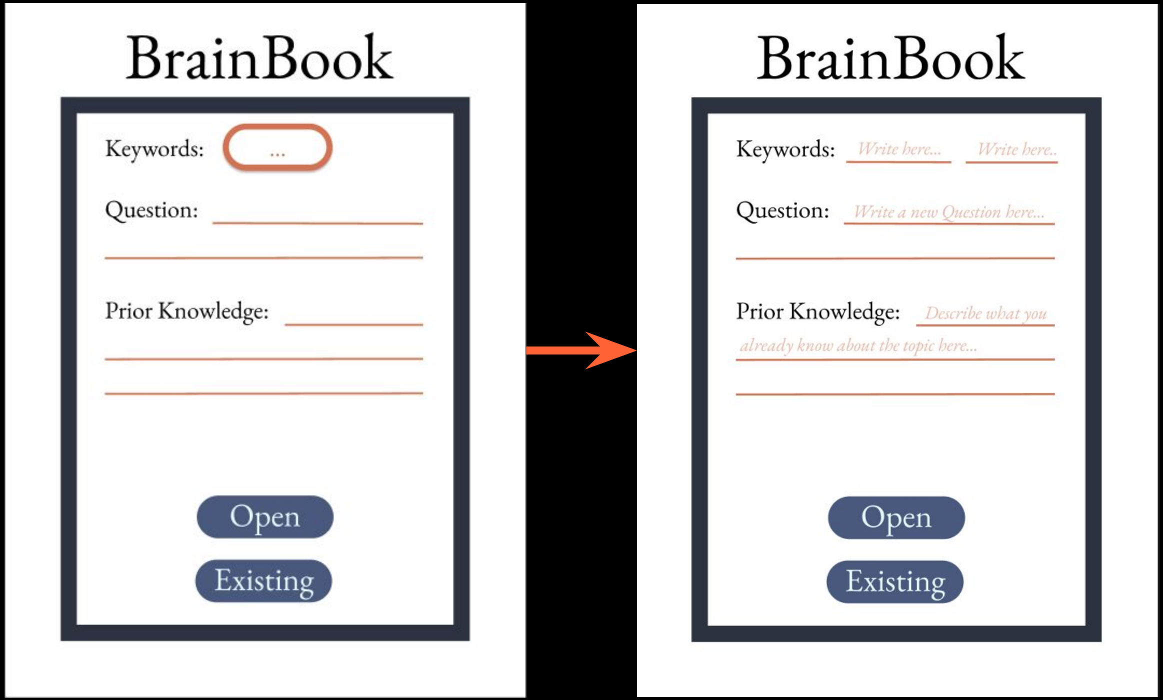


Redesign Cover Page

- We removed the pill and replaced it with an underline for overall consistency
- We added a second keyword slot to make it more obvious of what to write – short, single words
- We added placeholders to directly inform the user of what to write – no guesswork!

Usability Goal #1: this makes usage much more intuitive, improving the goal
- We will definitely use more placeholders going forward

This marginally contributes to **UG#2** by encouraging users to clearly state their question, which lead to better teaching down the line

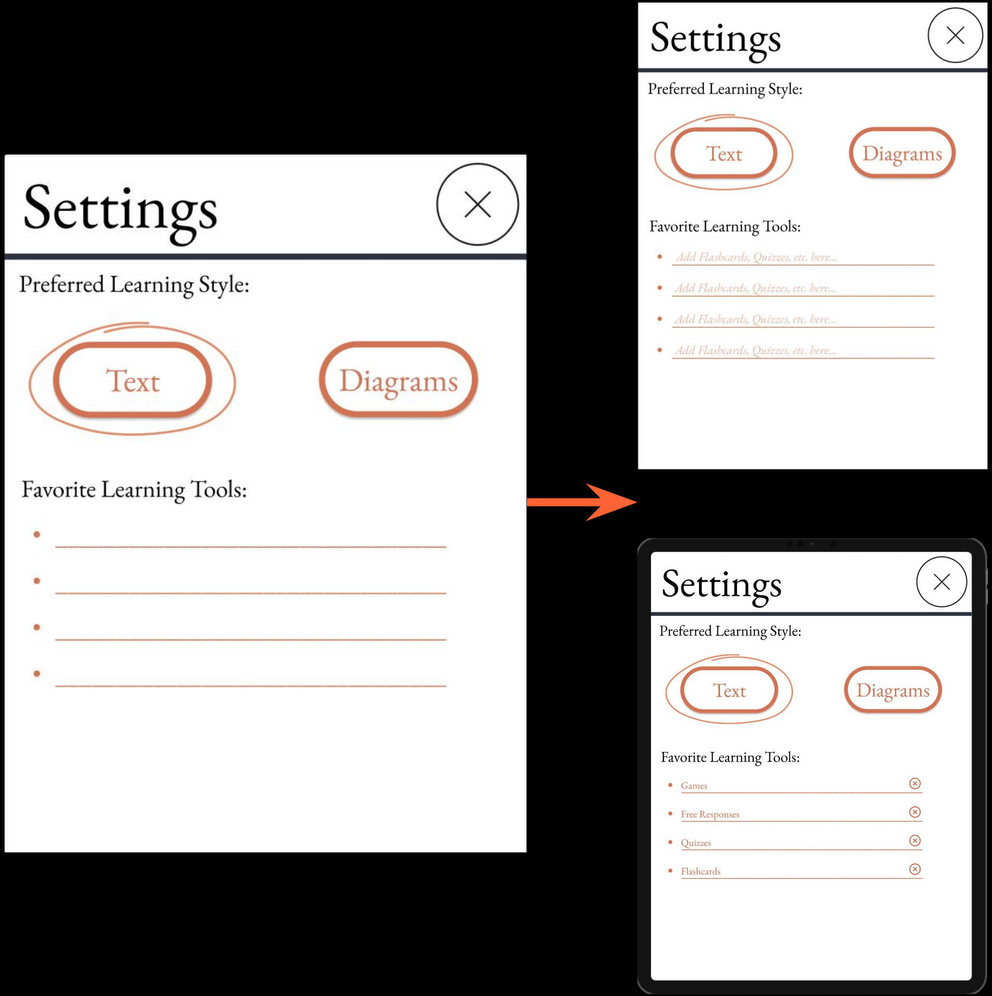


Modify Settings Page

- We added placeholders to inform the user what type of tools the system is looking for
    - No guessing!
- When a learning tool is specified, we added an X button to allow users to remove the learning tool
    - Necessary control

UG#1: this makes the UI far more intuitive, reducing usability errors

UG#2: better customization leads to better teaching, which ultimately leads to better teaching and fewer reasoning mistakes



Settings Saved Dialog

- This did not exist previously
    - Clicking X just returned to the previous page
- Now, this dialog tells the user that their settings are being saved
    - Reduce uncertainty

UG#1: this marginally improves user confidence, which contributes to this goal

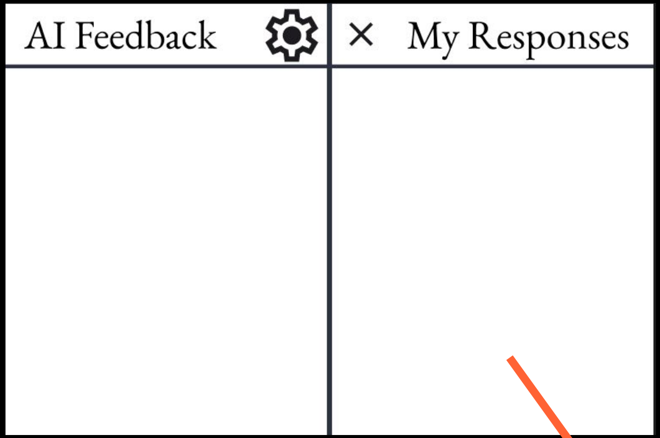
This does not contribute to **UG#2**

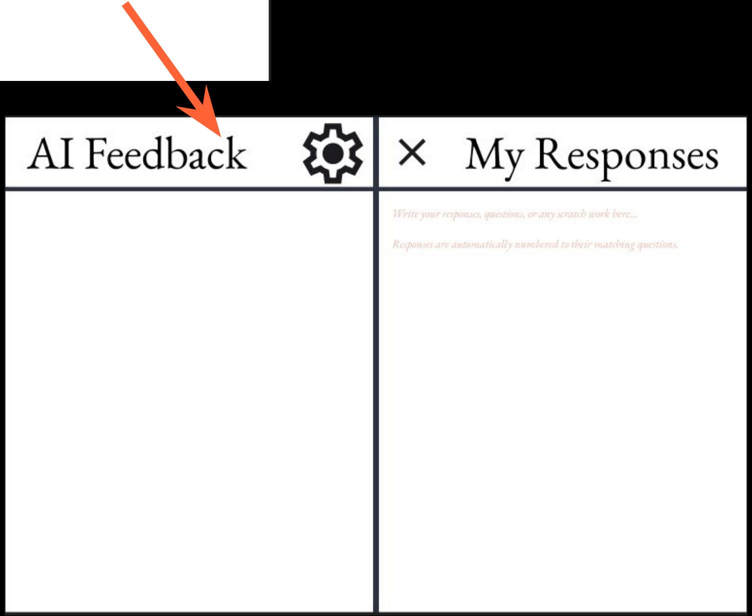
Add Placeholder

- A big issue was the user not being sure where to write initially
- We addressed this by adding a placeholder (when no user text has been written) that informs the user to write there

UG#1: This addresses a gap in the user's intuition and naturally guides them to where they need to go

This does not contribute to **UG#2**



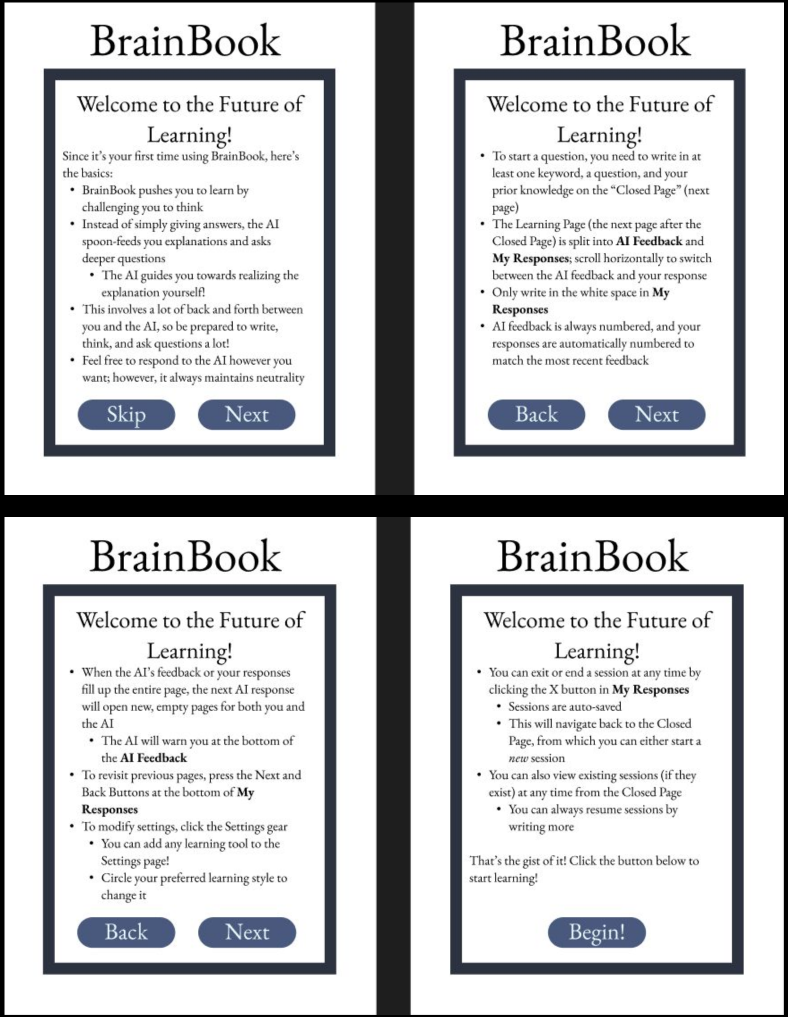


Onboarding Guide

- This did not exist previously
  - The App started at the Closed Page initially
- Now, the user is first told explicitly how to use the product completely, which takes a lot of guesswork out of the product

This contributes significantly to **UG#1** by teaching the user how to use the product. This reduces a significant amount of usability errors

This doesn’t really affect **UG#2**

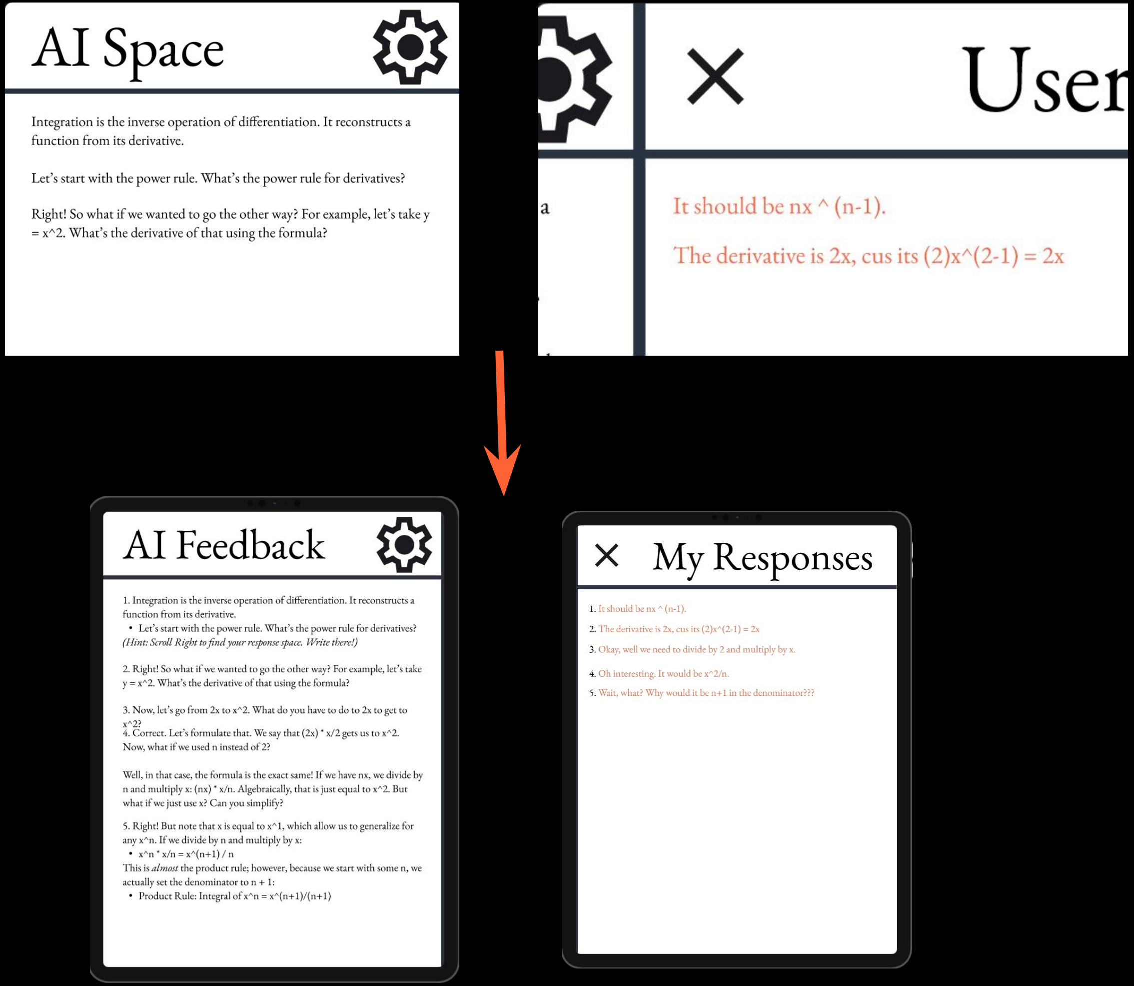


Numbering Feedback/Responses

- To avoid confusion between AI feedback and user responses (especially when browsing old sessions), matching feedback and responses are now numbered together
  - This ensures that users can directly link the two, reducing confusion
- The numbers are always in black to stand out

UG#1: This directly decreases usability issues by making it easier to dissect sessions

UG#2: This prevents any confusion between the user and the AI, preventing stupid mistakes

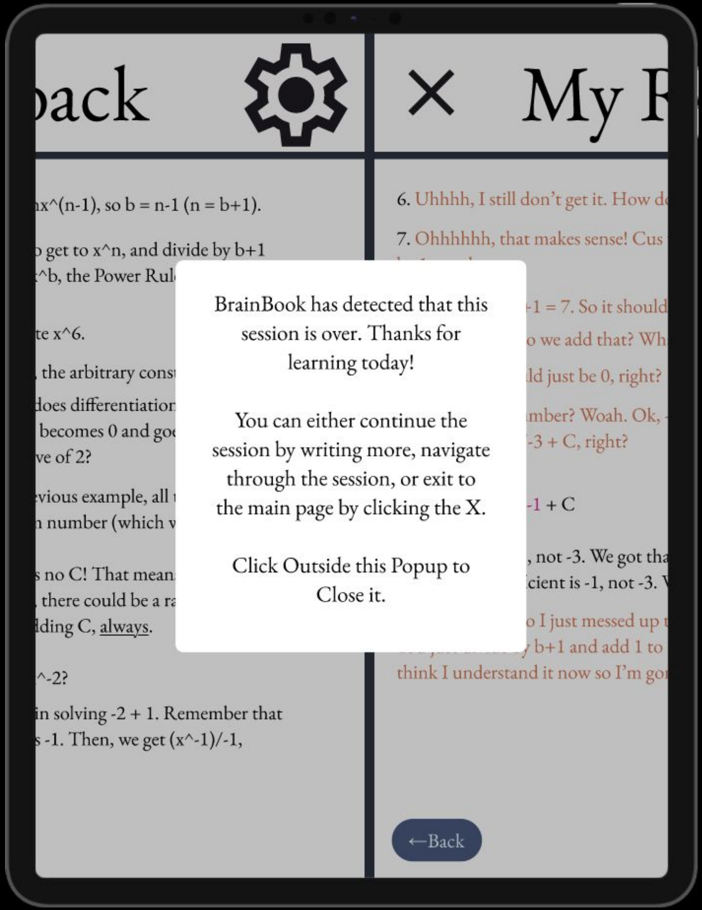


Session Completion Popup

- To make sure the user is fully aware that the session is over, the AI detects and informs the user what their options are after the session ‘ends’
- This reduces confusion over the session

UG#1: This marginally improves usability by reducing confusion

This does not affect **UG#2**



Indicate Page Reset/Control

- Before the page resets, the user informed by the AI
    - Removes any possible confusion, improving the learning experience
- The user can use the Next & Back buttons to navigate between pages
    - Gives the user the power to navigate the session

These changes improve **UG#1** and **UG#2** by reducing confusion and improving access to previous material, which boosts both usability and learning.

Thank you!

[Figma Prototype]